

Taeyun Roh

Seoul, Republic of Korea | txnrd.github.io | github.com/txnrd

Researcher in AI systems and multimodal reasoning with experience spanning agentic memory, efficient LLM inference, healthcare AI, and applied product development.

Education

Korea University <i>M.S. in Computer Science and Engineering</i>	Mar. 2026 – Present Seoul, Republic of Korea
Master’s student in Computer Science and Engineering.	
Korea University <i>B.S. in Computer Science and Engineering</i>	Mar. 2022 – Feb. 2026 Seoul, Republic of Korea
Undergraduate studies focused on systems, machine learning, and software engineering.	

Research Experience

Data Mining & Information Systems Laboratory, Korea University <i>Undergraduate Researcher, advised by Prof. Jae Woo Kang</i>	Mar. 2025 – Present Seoul, Republic of Korea
Research on agentic memory systems for AI agents.	
Network and Mobile System Laboratory, KAIST <i>Research Intern, advised by Prof. Sung-Ju Lee</i>	Jan. 2025 – Mar. 2025 Daejeon, Republic of Korea
Worked on efficient LLM inference, including on-device LLM inference.	
System Software Laboratory, Korea University <i>Undergraduate Researcher, advised by Prof. Gyeongsik Yang</i>	Jan. 2024 – Dec. 2024 Seoul, Republic of Korea
Researched federated learning for healthcare applications under resource heterogeneity.	

Selected Publications

CLAG: Adaptive Memory Organization via Agent-Driven Clustering for Small Language Model Agents <u>Taeyun Roh</u> , Wonjune Jang, Junha Jung, Jaewoo Kang	ACL 2026 Findings
Breaking Failure Cascades: Step-Aware Reinforcement Learning for Medical Multimodal Reasoning Junha Jung, Minbyul Jeong, Suhyeon Lim, Sungwook Jung, Jaehoon Yun, <u>Taeyun Roh</u> , Mujeen Sung, Jaewoo Kang	Under Review
When Choices Become Priors: Contrastive Decoding for Scientific Figure Multiple-Choice QA <u>Taeyun Roh</u> , Eun-yeong Jo, Wonjune Jang, Jaewoo Kang	Under Review

Industry & Development Experience

Aigensciences <i>AI Researcher</i>	Mar. 2025 – Aug. 2025 Seoul, Republic of Korea
- Built an LLM-based pipeline for automated hypothesis generation in drug discovery. - Developed a pediatric counseling chatbot designed for KakaoTalk integration.	
Blurting <i>Frontend Developer, Co-founder</i>	Jun. 2023 – Jan. 2025 Seoul, Republic of Korea
Co-founded a four-member startup and led end-to-end frontend development.	

- Developed and operated a blind-dating application that reached #73 in the Korea App Store Social Networking category.

Hiconsy
Data Analyst

Jan. 2022 – Jan. 2023
Seoul, Republic of Korea

- Built mathematical models for optimizing teaching assistant compensation.
- Applied regression analysis to study the relationship between student study time and academic performance.

Awards

- **1st Place, Korea University Capstone Design** (Prize: KRW 1,800,000) — Developed *NEWSQRAB*, an AI-driven system for automated news-to-short-form video generation using RAG-based LLMs.
- **2nd Place, Korea University NEXT Software Competition** (Prize: KRW 1,000,000) — Recognized as lead AI engineer for the *NEWSQRAB* project.
- **3rd Place, 2024 KIISE Conference (Undergraduate Category)** (Dec. 2024) — Awarded for the paper “*Analysis on Re-search for Data and Resource Heterogeneity in Split-Federated Learning.*”
- **2nd Place, 2023 Korea University Hackathon** (Prize: KRW 1,000,000) — Built Noonchi, a real-time visitor prediction and recommendation service for Han River parks.

Teaching Experience

Korea University
Teaching Assistant, Machine Learning (COSE 362), instructed by Prof. Jae Woo Kang

Fall 2025
Seoul, Republic of Korea

Academic Service

ACL 2026
Secondary Reviewer

2026

Selected Projects & Skills

Projects

- **Xalute** — Wearable health monitoring app developed with Korea University Anam Hospital for real-time health tracking and alerts.
- **Kakao Pediatric Response Chatbot** — LLM-based pedi-atric Q&A chatbot developed with Kakao Healthcare and Upstage.

Skills

- **Research:** Agentic memory, multimodal reasoning, LLM inference, healthcare AI, federated learning
- **Development:** Frontend development, applied LLM products, data analysis
- **Language:** English (TOEIC 940)